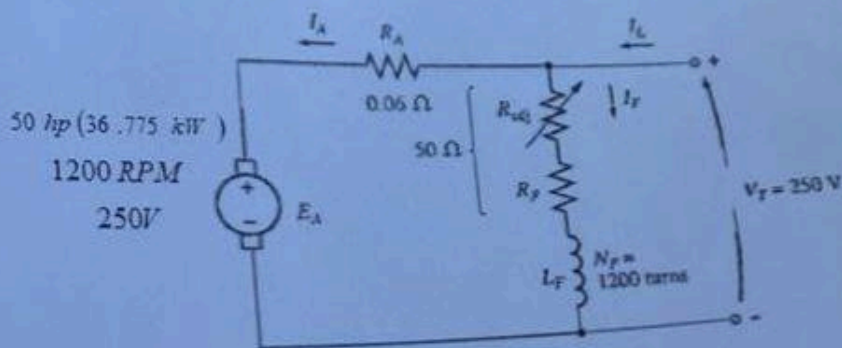


1.5 For self-excitation of a DC shunt generator, which condition is necessary?

- A) Field winding must have residual magnetism
- B) Field resistance > critical resistance
- C) Load must be connected before starting
- D) External DC supply is mandatory
- E) None of the above

Q2) Choose the correct answer:

The figure below shows the diagram of a  $50\text{hp}(36.775\text{kW})$ ,  $250\text{V}$ ,  $1200\text{RPM}$  dc shunt motor with compensating windings and an armature resistance of  $0.06\Omega$ . The field circuit of the motor has a total resistance  $R_{adj} + R_F = 50\Omega$ , which produces a no-load speed of  $1200\text{RPM}$ . There are 1200 turns per pole on the shunt field winding.



16. The rated current of this motor is:

- A) 147.1A
- B) 199.5A
- C) 248.6A
- D) 575.5A
- E) None of the above

17. The rated torque of this motor is:

- A) 585.3 N.m
- B) 292.7 N.m
- C) 379.5 N.m
- D) 146.3 N.m
- E) None of the above

18. The speed of this motor when the line current is 100A:

- A) 987 RPM
- B) 1173 RPM
- C) 1157.6 RPM
- D) 1020 RPM
- E) None of the above

19. The speed of this motor at rated current is:

- A) 987 RPM
- B) 1020 RPM
- C) 1173 RPM
- D) 1157.6 RPM
- E) None of the above

20. The speed regulation of this motor is:

- A) 2.78%
- B) 6.75%
- C) 5.67%
- D) 3.66%
- E) None of the above

Q3) Solve the following problem for simplex progressive Lab-winding: 10 points

A six pole DC generator has a flux per pole of  $30\text{mWb}$ . The armature has 536 conductors connected as a lap. The DC generator runs at  $1050\text{rpm}$ , and delivers a rated armature current of  $225\text{A}$  to a load connected to its terminals. The field and armature windings are connected in series ( $R_f = R_a = 0.05\Omega$ ). Combined friction, windage, and core losses are  $2\text{kW}$ . Calculate:

- 1) Generated voltage,  $E_g$
- 2) The power developed by armature,  $P_{dev}$
- 3) The developed torque,  $T_{dev}$
- 4) Terminal voltage,  $V_t$
- 5) Efficiency



Solve showing your work (10 Points)

A separately excited DC motor is rotated at 1000rpm, The variation of armature terminal voltage as a function of field current is measured under no-load conditions and table below:

|       |     |     |     |     |     |     |     |     |     |
|-------|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| $I_F$ | 0.0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 |
| $V_T$ | 0   | 30  | 60  | 85  | 102 | 115 | 124 | 130 | 134 |

The field winding supply 24V and the field resistance is adjustable. The armature winding resistance  $0.2\Omega$  and the armature terminal voltage is 130V.

- Calculate the field current if the motor is operated with no-load at 1000 rpm
- The motor drives a load at 1200 rpm. Calculate the armature voltage at 1200 rpm if the field resistance  $60\Omega$
- Calculate the torque for the above condition

$I_F$  .....

$E_a$  = .....

$T$  = .....



Q1 Choose the correct answer.

1. The full-load copper loss of a transformer is 4000W. At half-load, the copper loss will be:

A) 1000W      B) 2000W      C) 1000W      D) 500W

2. A transformer has a primary winding having 100 turns, secondary winding having 500 turns.

- [illegible]



Solve showing your work (10 Points)

A DC motor delivers 15 kW at 1000 rpm. Total losses are 2.5 kW.

Find:

- a) Input power
- b) Efficiency
- c) Developed torque

$P_{in} = \dots\dots\dots$

$\eta = \dots\dots\dots$

$T = \dots\dots\dots$



correct answer:

A constant current flows in a horizontal wire in the plane of the paper from east to west. The point at which the direction of magnetic field is from North to South is:  
 directly above the wire  
 directly below the wire  
 one of the above  
 B) In the plane of the paper, in the north side of the wire  
 D) In the plane of the paper, in the south side of the wire

The full-load copper loss of a transformer is 2000W. At half-load, the copper loss will be:  
 1000W B) 2000W C) 1000W D) 500W E) None of the above

A transformer with primary winding having 100 turns, secondary winding having 500 turns, if 200 A is flowing into the primary winding, the current out of the secondary winding will be:  
 40A B) 100A C) 500A D) 1000A E) None of the above

The torque in a DC shunt motor is proportional to:

A) Armature current B) Field current only  
 C) Square of armature current D) Armature resistance E) None of the above

If the field winding of a DC shunt motor opens during operation, the motor will:

A) Stop immediately B) Run at dangerously high speed  
 C) Run at constant speed D) Run at half speed E) None of the above

The most common method of speed control in a DC shunt motor is:

A) Armature resistance control B) Field flux control  
 C) Voltage control D) Scott-T connection E) None of the above

If the work done by a motor is 1000J while its torque is 10Nm, the number of complete revolution done by the shaft of this motor is:

A)  $100\pi$  B)  $50\pi$  C)  $\frac{50}{\pi}$  D)  $\frac{100}{\pi}$  E) None of the above

To reduce sparking, DC machines are provided with:

A) Dummy coils B) Interpoles (commutating poles)  
 C) Extra insulation between coils D) Series resistance in brushes  
 E) Compensating winding F) None of the above

If the field winding, of a lightly loaded DC shunt motor, gets opened, the:

A) motor will stop B) motor will decrease slightly  
 C) motor speed will increase slightly D) motor will pick up very high speed  
 E) None of the above

10. If a DC motor has a full-load speed of 750 RPM and speed-regulation of 10%, the no-load speed of this motor is:

A) 825 RPM B) 800 RPM C) 700 RPM D) 675 RPM E) None of the above

11. In the separately excited DC motor, the relation between the motor speed,  $n$ , and the field current,  $I_f$ , is:

A)  $n$  is directly proportional to  $I_f$   
 B)  $n$  is directly proportional to the square of  $I_f$   
 C)  $n$  is inversely proportional to  $I_f$   
 D)  $n$  is inversely proportional to the square of  $I_f$   
 E) None of the above

12. The speed-torque characteristic of a DC series motor is:

A) Constant speed B) Nearly constant torque  
 C) High starting torque, speed decreases with load D) Linear between speed and torque  
 E) None of the above

13. In the DC series machine, the magnetic flux is proportional to the:

A) square of the armature current B) armature current  
 C) square root of the armature current D) cube of the armature current E) None of the above

14. If the field winding of a DC shunt generator is short-circuited, the output voltage will:

A) Increase suddenly B) Remain the same  
 C) Drop to zero D) Reverse in polarity E) None of the above



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**Q3) Solve showing your work (10 Points)**

A 240-V, 100-A DC shunt motor. This motor has an armature resistance of 0.14 ohm, field resistance of 200 ohm, and 1500 turns. A resistance with range from 0 to 300 ohm. With compensating windings and interpoles and field turn of 1500 turns. A magnetization curve for this motor at 3000 r/min can be summarized in the following table

| $I_f$ | 0.1 | 0.3 | 0.48 | 0.5 | 0.6 | 0.75 | 1.1 | 1.2 | 1.3 |
|-------|-----|-----|------|-----|-----|------|-----|-----|-----|
| $E_a$ | 60  | 140 | 195  | 200 | 220 | 245  | 278 | 282 | 285 |

- What is the no-load speed of this motor when  $R_{adj} = 120$  ohm?
- What is its full-load speed?
- What is its speed regulation?

$n = \dots\dots\dots$

$E_a = \dots\dots\dots$

$SR = \dots\dots\dots$